An effective QUBO formulation for knapsack problems

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2025 IEEE International Conference on Quantum Artificial Intelligence (QAI) | 979-8-3315-6986-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/QAI63978.2025.00039

***Abstract*—Combinatorial optimization problems are central to addressing complex decision-making challenges in finance and represent a promising avenue for demonstrating quantum advantage using near-term quantum hardware. While significant progress has been made in encoding unconstrained optimization problems into Hamiltonian formulations for quantum computing, constrained optimization problems remain an open challenge. The knapsack problem, a class of NP-hard constrained combinatorial optimization problems, exemplifies this issue. Existing Hamiltonian encodings for knapsack problems typically employ either a prohibitive number of slack variables, exceeding the qubit capacity of current quantum devices, or unbalanced penalty methods that demand extensive parameter tuning to achieve the optimal solutions. This paper introduces a novel Hamiltonian formulation that addresses these limitations by requiring only two slack variables and significantly reducing the need for penalty parameter calibration. Experiments conducted on a quantum annealer for small-scale instances demonstrate the proposed method's superior performance compared to existing approaches, achieving higher success rates in locating the optimal solutions while maintaining hardware feasibility. The results highlight the potential of this formulation to advance the practical application of quantum optimization to constrained combinatorial problems in finance and beyond.**

I. Introduction

As quantum computing evolves from theoretical promise to practical utility, the financial sector stands at a critical juncture. Institutions that fail to adapt risk obsolescence; those that innovate may gain unprecedented competitive edges.

The financial industry is inherently driven by optimization. Whether allocating capital, managing risk, or constructing portfolios, decision-makers perpetually face the challenge of maximizing value under constraints.

A. Knapsack problems in financial sector

Among the most pervasive frameworks for such decisions is the knapsack problem, a combinatorial optimization model that underpins critical financial workflows. At its core, the knapsack problem encapsulates the trade-off between value and scarcity: given a set of items with associated weights and values, the goal is to select a subset that maximizes total value without exceeding a predefined weight capacity. This optimization problem is classified as NP-hard [17]. Although conceptually straightforward, its computational complexity and adaptability to real-world constraints have made it indispensable and notoriously challenging in financial applications.

In finance, the knapsack problem manifests itself in various forms. Consider portfolio optimization, where investors aim to select assets that maximize returns while adhering to budget limits or risk thresholds. Here, each "item" represents an investment opportunity with a monetary cost (weight) and expected return (value) [31]. Similarly, in capital budgeting, firms must choose between projects or acquisitions under finite capital reserves, balancing upfront costs against long-term profitability. Regulatory constraints, such as sector exposure limits or liquidity requirements, further transform these problems into multidimensional knapsack variants [37]. Even risk management adopts knapsack-like logic, where institutions prioritize hedging strategies or asset diversification to "fill" a risk tolerance "capacity" while minimizing potential losses [16]. These applications share a common thread: they require binary decision-making under rigid constraints.

Classical methods for solving knapsack problems, such as dynamic programming [1], [3], branch-and-bound [6], integer linear programming (ILP) [36], or a hybrid of these approaches [28], [22] struggle with these complexities. Although effective for small-scale instances, their computational costs grow exponentially with problem size, rendering them impractical for large financial data sets or real-time decision making. For example, a portfolio manager evaluating thousands of assets with correlated returns cannot feasibly enumerate all combinations, while a bank allocating capital across global divisions faces NP-hard complexity in multidimensional constraints.

As an alternative, new computational paradigms such as quantum technology are being explored to achieve speed-ups and accuracies over traditional methods [8], [34]. Quantum computing enables the solution of certain types of complex problems significantly faster than classical algorithms [13], [33].

B. Solving knapsack problem using QUBO

Research aimed at solving the knapsack problem using quantum computing is still in early stages. A quantum genetic algorithm was used to solve knapsack problem in [2]. A quantum inspired cuckoo search was proposed in [19]. Variational quantum algorithm was used in [4]. The most popular approach to address the combinatorial optimization problems using quantum computers is to convert the problem

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into a Quadratic Binary Optimization Problem (QUBO) [18] of the form

$$H = \sum_{ij} Q_{ij} x_i x_j,$$

where Q_{ij} is an element of a symmetric matrix and x_i is a binary variable. The minimum of the objective function, i.e., the Hamiltonian H is the solution to the optimization problem. There are two distinct paradigms of quantum computing, namely 'gate-based quantum computing' and 'quantum annealing'. In a gate-based quantum computer, a sequence of instructions in the form of quantum gates are applied to the initial quantum state. Then the final state of the evolved quantum system is determined by measuring some, or all, of the qubits in the evolved quntum state. On the other hand, a quantum annealer allows the quantum system to evolve on its own following one of the fundamental equations of quantum mechanics. Specifically, the Schrödinger equation. In the end, the system reaches its ground state or minimum energy state. Different algorithms exist to solve QUBO using a gate-based quantum computer, e.g., Variational Quantum Eigensolver (VQE) [27], Quantum Approximation Optimization Algorithm (QAOA) [9]. One major limitation of gate-based quantum computers is the noise that gets accumulated with the application of quantum gates, making the result unreliable. Since quantum annealers do not require quantum gates, the annealing process is less affected by noise. Several approaches for solving QUBO using QAOA and QA can be found in literature, e.g, in supply chain management [15], finance [26], [24], [10], energy [32], network [35], automotive [21], traffic signal [14], among many others.

Among various kinds of combinatorial optimization problems, those with inequality constraints such as knapsack problems are more challenging to encode into QUBO formalism. One popular approach is to introduce slack variables [20]. The number of slack variables can grow quickly depending on the problem instance. Too many slack variables often disrupt the energy landscape and make it difficult for the quantum hardware to find the optimal solution. It was shown that quantum computers were unable to find the optimal solutions even for very small instances using such formalisms [29], [30]. There was an alternative approach called 'unbalanced penalization' [23]. Even though this approach was more successful in finding the optimal solution than the previous method, the optimal solution does not always coincide with the ground state of the Hamiltonian. As a result, the penalty constants need a large amount of fine-tuning.

In this work, we propose a new QUBO formalism that requires only two slack variables to impose the inequality constraint of the knapsack problem. We also provide bounds for the penalty constants so that within those bounds quantum hardware can find the optimal solutions by solving the QUBO. We test our QUBO formalism on a set of small knapsack instances using quantum annealing. We found that our formulation was able to obtain the optimal solution for most of the instances with minimal tuning. We also compare the performance with other existing formulations and showcase the improvements from our approach.

II. Review of existing QUBO formulations

Let us consider the following knapsack problem with N items and knapsack capacity $\mathcal{W}$. Weight of each item is denoted by w_i. Profit associated with each item is denoted by v_i. The goal is to maximize the profit while respecting the knapsack capacity. Mathematically the knapsack problem can be expressed as follows:

$$\text{maximize } \sum_{i=1}^{N} v_i x_i, \tag{1}$$

$$\text{such that } \sum_{i=1}^{N} w_i x_i \leq \mathcal{W}, \tag{2}$$

where $x_i = \{0, 1\}$, is the binary variable associated with item i. If $x_i = 1$, the item i goes into the knapsack, and if $x_i = 0$, it does not. This class of knapsack problem is known as one-dimensional $0/1$-knapsack problem. It was shown in [29] that quantum annealers failed to reach the optimal solution for such knapsack problems. A QUBO formulation proposed by Lucas in [20] was used in article [29] to show the inefficiency of a quantum annealer in solving knapsack problem. We call the formulation proposed in [20] as "QUBO-type-I". This formulation can be summarized as follows.

A. QUBO-type-I:

The QUBO hamiltonian is given by $H = H_A + H_B$ with

$$H_A = A\left(1 - \sum_{\alpha=1}^{\mathcal{W}} y_\alpha\right) + A\left(\sum_{\alpha=1}^{\mathcal{W}} \alpha y_\alpha - \sum_{i=1}^{N} w_i x_i\right)^2, \tag{3}$$

where y_α, with $1 \leq \alpha \leq \mathcal{W}$ is a binary slack variable. Only one of the slack variables assumes non-zero value. y_α takes value 1 if the final weight of knapsack is α. The first part of H_A enforces that the weight can take only one value and the second part enforces that the total weight of the objects in the knapsack equals the value α. Finally,

$$H_B = -B \sum_{i=1}^{N} v_i x_i \ . \tag{4}$$

The constants A and B have to be chosen such that $0 < B \max(v_i) < A$, so that adding one item to the knapsack, which makes it too heavy, is not allowed.

B. QUBO-type-II:

It was shown in [30] that QUBO-type-I formulation does not always obey the knapsack constraints. The authors of [30] proposed a modification to H_A of QUBO-type-I. We call the modified formulation as "QUBO-type-II". According to this modified formulation, the QUBO Hamiltonian is given by $H = H_A + H_B$, where $H_B = -B \sum_{i=1}^{N} v_i x_i$, and

$$H_A = A_1\left(1 - \sum_{\alpha=1}^{\mathcal{W}} y_\alpha\right) + A_2\left(\sum_{\alpha=1}^{\mathcal{W}} \alpha y_\alpha - \sum_{i=1}^{N} w_i x_i\right)^2,$$

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where $A_1 \geq \sum_i v_i$, and $A_2 \geq \max\{v_i : i \in N\}$.

The number of qubits required for both QUBO-type-I and QUBO-type-II is equal to $N + \mathcal{W}$.

C. QUBO-type-III:

Another variation of QUBO-type-I is widely used in the literature [7], [12], [30], [11]. We call this variation "QUBO-type-III". The number of qubits required was significantly reduced in this formulation. H_A from Eqn. (3) is modified as follows:

$$H_A = A\left(\sum_{j=0}^{M-1} 2^j y_j + (\mathcal{W} + 1 - 2^M) y_M - \sum_{i=1}^{N} w_i x_i\right)^2,$$

where $M = \lceil \log_2 \mathcal{W} \rceil$. Now there are $M+1$ number of slack variables $y_0, y_1, \ldots y_M$, and multiple slack variables can be non-zero. The total number of qubits required is $N + M + 1$.

Several variations of the above three QUBO formulations have been considered in [5]. We verified that they perform almost identical to their original versions.

D. QUBO-type-IV:

An unbalanced penalization approach without the slack variables was proposed in [23]. The QUBO Hamiltonian for the knapsack problem in this approach takes the following form:

$$H = -\sum_{i=1}^{N} v_i x_i + A\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)^2 + B\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right) . \quad (5)$$

This formulation was also investigated in [10]. The main advantage of this formulation is that the number of qubits required is significantly reduced as there is no slack variable present. However, the main drawback is that inequality constraints are not strictly enforced and hence the ground state of the Hamiltonian may not provide the optimal solution.

III. Proposal for a new QUBO formalism

We propose a new QUBO formalism that uses minimal qubits while ensuring that the knapsack constraints are met. We call this new formulation "QUBO-type-V". The QUBO Hamiltonian is given by

$$H = A\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)^2 - B(y_0 + y_1)\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right) + C(y_0 + y_1 - 1)^2 - \sum_{i=1}^{N} v_i x_i, \quad (6)$$

where y_0 and y_1 are two binary slack variables. Let us discuss the role of each term of Eqn. (6) starting with the last term.

1) $-\sum_{i=1}^{N} v_i x_i$: This negative sign in this term ensures that the profit is maximized when we minimize the Hamiltonian.
2) $C(y_0+y_1-1)^2$: We set the constant C very large so that this term ensures one of the slack variables is non-zero.
3) $B(y_0 + y_1)\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)$: When one of the slack variables is non-zero, this term enforces the total weight of selected items remains within the knapsack capacity. When knapsack capacity is obeyed, i.e., $\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)$ has a positive sign, this term rewards the minimization of the Hamiltonian. On the other hand, when the knapsack capacity is violated, the sign becomes negative, and it penalizes the minimization. To endure that this linear term does not overpower the quadratic penalty from the second term, we require

$$\left| B\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)\right| < A\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)^2 . \quad (7)$$

This constraint gives us the following bound on the constant B:

$$B < A\left|\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)\right| . \quad (8)$$

Let's upper bound it at worst case, i.e., $B < A\Delta W$, where $\Delta W = \max(\mathcal{W}, \sum_{i=1}^{N} w_i x_i - \mathcal{W})$.

4) $A\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)^2$: We want the penalty for violating weight constraints to dominate any gain from the profit term, i.e., for any infeasible solution, the total energy should be higher than the feasible ones. The worst-case profit gain is $\sum_i v_i \leq N V_{\text{max}}$, where V_{max} is the maximum profit from an item. Maximum possible weight deviation is $\left(\mathcal{W} - \sum_{i=1}^{N} w_i x_i\right)^2 \leq \Delta W^2$. This imposes a lower bound on A, i.e.,

$$A \geq \frac{N V_{\text{max}}}{\Delta W^2} . \quad (9)$$

Instances	Number of items (N)	Capacity ($\mathcal{W}$)	Optimal profit	Number of slack variables required by QUBO				
				type-I	type-II	type-III	type-IV	type-V
A	10	269	295	269	269	9	0	2
B	20	878	1024	878	878	11	0	2
C	4	20	35	20	20	6	0	2
D	4	11	23	11	11	5	0	2
E	15	375	481.0694	375	375	10	0	2
F	10	60	52	60	60	7	0	2
G	7	50	107	50	50	7	0	2
H	23	10000	9767	10000	10000	14	0	2
I	5	80	130	80	80	7	0	2
J	20	879	1024	879	879	11	0	2

TABLE I: Description of instances that we use for our experiments, and the number of slack variables required to solve QUBO for each formalism.

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IV. Method

A. Quantum Processor

We perform our experiments on a quantum annealer. We use quantum hardware with 5612 active qubits, and 15 qubits connectivity, i.e., at most 15 qubits are connected with each other. The largest clique that can be embedded on this hardware has 180 nodes. The node count varies depending on the connectivity of the problem Hamiltonian. The hardware is connected using a Pegasus graph structure. It is important to specify the mapping of the problem graph to the hardware topology in order to solve it. We use `EmbeddingComposite` for embedding our QUBO Hamiltonian to the `Advantage` hardware.

B. Problem Instances

We use small instances of $0/1$-knapsack problems from an online publicly available database [25]. The knapsack problem instances are listed in Table I. The instances are small so that it is possible to solve them exactly using classical solvers.

We have described a total of five QUBO formulations including our new proposed one in the previous two sections. Different QUBO formulations use different number of qubits. The number of qubits required for each formulation are listed in Table II. Due to the hardware limitations, we can only implement a subset of the above instances for some formulations depending on the required number of qubits.

QUBO type	Qubits required	Instances implemented
I	$N + \mathcal{W}$	C,D,F,G,I
II	$N + \mathcal{W}$	C,D,F,G,I
III	$N + \lceil \log_2 \mathcal{W} \rceil + 1$	All
IV	N	All
V(New)	$N + 2$	All

TABLE II: Number of qubits required for different QUBO formulations and list of knapsack instances that can be implemented in the hardware. N and $\mathcal{W}$ are listed in Table I.

V. Results

We summarize the primary result of our experiment in Table III. We put a "✓" when quantum annealer was able to achieve the optimal result, a "✗" when quantum annealer did not reach the optimal solution, and "NA", i.e., "Not Available" when not enough qubits are available for the QUBO implementation. It is clear from Table III that QUBO-type-IV and QUBO-type-V performs better than the rest of the QUBO formulations. As explained in [23], [10], QUBO-type-IV or the unbalanced QUBO Hamiltonian ground state does not always overlap with the optimal solution. Even though this formalism reached the optimal solution in quantum annealer for most of the instances that we considered, it required heavy fine-tuning for the penalty constants A and B. In contrast, our proposed QUBO-type-V was able to achieve the optimal solution with minimal tuning of the constants A and B. Fig. 1 shows the deviation of the ground state from the optimal solution for knapsack instance F as a function of the penalty constants. As we can see from the plots that QUBO-type-V

Instances	I	II	III	IV	V
A	NA	NA	✗	✓	✓
B	NA	NA	✗	✓	✓
C	✗	✗	✓	✓	✓
D	✗	✗	✗	✓	✓
E	NA	NA	✓	✓	✓
F	✗	✗	✗	✓	✓
G	✗	✗	✓	✓	✓
H	NA	NA	✗	✗	✗
I	✗	✗	✓	✓	✓
J	NA	NA	✗	✓	✓

TABLE III: This table shows performances of different QUBO formulations. A ✓denotes that QUBO was able to reach the optimal value, ✗means the optimal value was not obtained. NA means the instances were too large for hardware implementation using that QUBO formulation, hence the optimization result was not available.

was able to achieve the optimal result more frequently than QUBO-type-IV. We use simulated annealing to compute the ground state of the QUBO Hamiltonian. Since the instances we use only require a small number of qubits (10 and 12 for type-IV and type-V formulations respectively for instance F), simulated annealing is a good enough approximation of the corresponding quantum annealing. We do similar analysis for

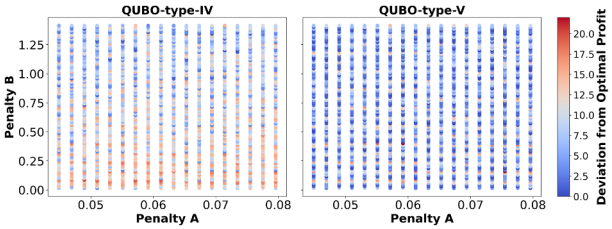

Fig. 1: Plot of the difference between the ground state solution and the optimal solution as a function of penalty constant A and B. Left panel is for QUBO-type-IV and the right panel is for QUBO-type-V. The dark blue color indicates that the ground state of the Hamiltonian overlaps with the optimal value of the knapsack instance, and red indicates large deviations. This plot is generated for knapsack instance F using simulated annealing.

other instances from Table I and observe a similar pattern. We include those plots in Appendix A.

Then we randomly chose any pair of the optimal penalty constants from the simulated annealing result and implement those in quantum annealer. We were able to obtain the optimal or a near-optimal result for most of the cases from quantum annealer.

VI. Discussions

We present a new approach to encode the inequality constraint of combinatorial optimization problem into QUBO formalism. We tested it against existing QUBO formulations. Both QAOA and QA can be used to solve QUBO. Quantum annealers(QA) felt more natural choice to us as they are

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specially built for solving such problems. Our method demonstrates competitive and often superior performance in terms of solution quality and convergence. One of the main motivations behind the new approach was to capture the inherent structure of knapsack constraints without requiring a large number of additional variables. As we have seen in the earlier sections, the existing formulations often rely on complex encoding strategies requiring a large number of slack variables. Our formalism requires only two slack variables thus reducing the number of qubits required. This is extremely beneficial for near-term quantum computing devices with limited number of qubits. There's only one other approach that required less qubits than ours, but that approach was not robust with respect to the choice of penalty terms. One needs to fine tune carefully to achieve the optimal solution. Our approach exhibits better robustness with respect to the choice of penalty parameters reducing the need of extensive parameter tuning.

While results are promising, it is important to mention that we have considered only small knapsack instances due to quantum hardware limitations. We hope our approach will continue to provide better solutions for larger knapsack instances as hardware quality improves. The one immediate generalization would be to apply this approach to multiple knapsack problems. Whether a similar approach can provide competitive solutions to other constrained optimization problems such as traveling salesperson, bin packing, vehicle routing is also worth investigating.

Appendix

We have shown how our QUBO proposal i.e., QUBO-type-V compare with QUBO-type-IV in terms of finding the optimal solution for instance F in the main text Fig. 1. We have tested a total of ten knapsack instances as listed in Table I. Here we show the comparison for rest of the nine instances. For the sake of space, we split them into two plots. Fig. 2 shows plots for instance A, B, C, D and E. Fig. 3 shows plots for instances G, H, I, and J. It is evident from the plots that QUBO-type-V consistently performs better than QUBO-type-IV. The reason of horizontal trend in the scatter plot for instance B and instance H is that values of penalty A (horizontal axis) are taken more closed spaced compared to the values of penalty B (vertical axis).

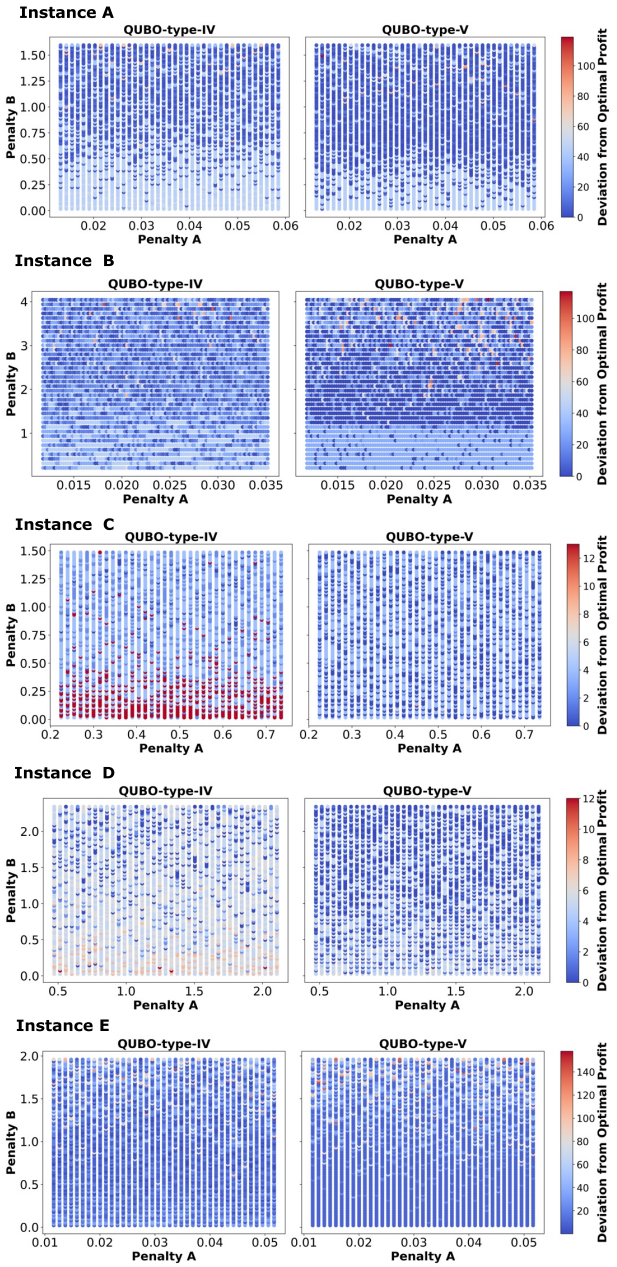

Fig. 2: Plot of deviation of ground state from the optimal solution as a function of A and B for Instance A,B,C,D,E. Left panel shows plots for QUBO-type-IV and right panel is for QUBO-type-V

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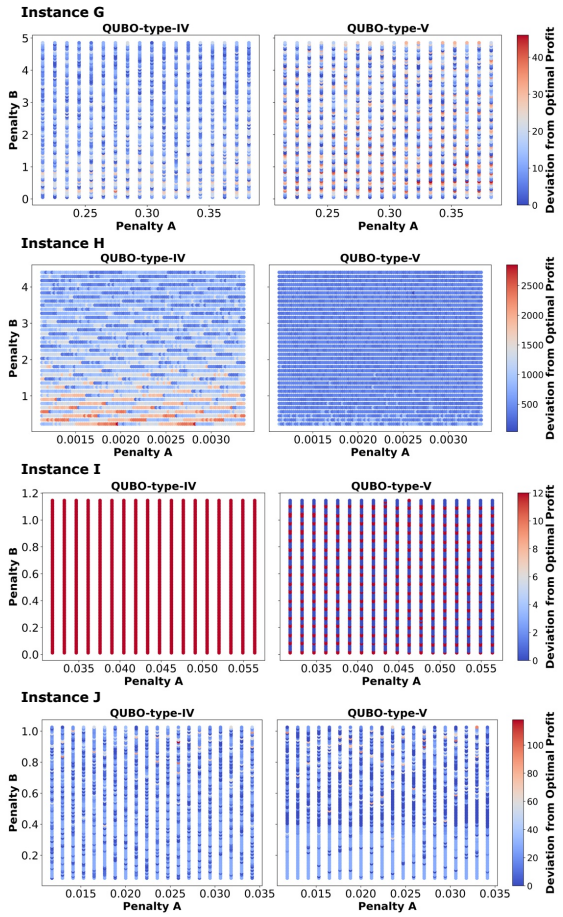

Fig. 3: Plots for instances G,H,I,J

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